

# Temperature & Humidity Monitoring System

Real-Time Environmental Sensing using Arduino Uno and DHT11 Sensor

## 1. Project Overview

This project implements a real-time temperature and humidity monitoring system using Arduino Uno and the DHT11 sensor. Environmental readings are collected every two seconds and displayed through both the Serial Monitor and a 16x2 LCD display. The project demonstrates the fundamental architecture of IoT systems including sensor acquisition, edge processing, real-time monitoring, and human-readable output interfaces.

## 2. Components Required

| Component    | Specification          | Qty |
|--------------|------------------------|-----|
| Arduino Uno  | ATmega328P Board       | 1   |
| DHT11 Sensor | Temperature & Humidity | 1   |
| 16x2 LCD     | I2C Module             | 1   |
| Breadboard   | 830 Point              | 1   |
| Jumper Wires | Mixed                  | 15  |

## 3. Circuit Diagram

+5V ----- DHT11 VCC  
GND ----- DHT11 GND  
D2 ----- DHT11 DATA

A4 (SDA) ----- LCD SDA  
A5 (SCL) ----- LCD SCL

## 4. Arduino Source Code

```
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);
```



## 5. Code Explanation

### Libraries

The program uses DHT, Wire, and LiquidCrystal\_I2C libraries for sensor communication and LCD control.

### Setup Function

Initializes serial communication, LCD display, and DHT11 sensor.

### Loop Function

Reads temperature and humidity continuously and updates the LCD and Serial Monitor.

### Error Handling

Checks for failed sensor readings using `isnan()` and displays an error message.

## 6. Expected Output

Temperature: 27.4 C  
Humidity: 63 %

Temperature: 27.5 C  
Humidity: 64 %

## 7. Future Enhancements

- Cloud integration using MQTT and ESP32
- Real-time dashboard visualization
- Mobile app notifications
- Data logging using SD card
- AI-based temperature prediction

## 8. Conclusion

This project demonstrates the core architecture of IoT monitoring systems using affordable hardware. The prototype can be expanded into larger smart-home, industrial, or environmental monitoring systems.